

THE MULTI-AGENT COGNITIVE SUBSTRATE THEOREM (MACS)

*A Structural Law of Network-Distributed Epistemic Behavior, Visibility Divergence, and
Cross-Agent Stability*

Author: SignalRupture

PREFACE

The Multi-Agent Cognitive Substrate Theorem (MACS) completes the SignalRupture (SR) Substrate Trilogy. CRI concerns persistent observation. AIS concerns delegated institutional judgment. MACS concerns the condition that follows when humans, institutions, computational models, autonomous agents, and other decision-relevant systems repeatedly observe, classify, interpret, and alter one another's outputs.

MACS is not a theory of artificial consciousness. It does not claim that machines think like humans, that cognition literally merges into one collective mind, or that current societies have already entered a fully network-distributed cognitive substrate. Its use of cognition is functional and relational. A participant qualifies when its outputs materially alter what another participant observes, classifies, prioritizes, decides, or learns.

The core proposition is relational and empirically bounded. When epistemic and decision functions become recursively distributed across heterogeneous participants, system stability can no longer be explained only by the accuracy or reliability of each participant in isolation. Interaction structure itself becomes causally important.

A human can be individually rational. A model can perform adequately on its benchmark. An institution can possess legitimate procedures. The combined network can still generate instability if outputs are misaligned, asynchronous, recursively amplified, differently grounded, or difficult to reconcile across layers.

MACS therefore moves the unit of analysis from the isolated agent to the coupled network.

Its empirical basis is presently the most prospective of the trilogy. Distributed cognition research has long treated cognition as extending across people, artifacts, and environments rather than remaining exclusively inside individuals (Hutchins, 1995; Hollan, Hutchins, & Kirsh, 2000). Contemporary human-AI collaboration research likewise studies conditions under which AI systems become teammates, decision partners, or components of hybrid intelligence rather than passive tools (Dellermann et al., 2019; Seeber et al., 2020). Research on machine behaviour further emphasizes that AI systems increasingly mediate social, economic, cultural, and political interactions and

therefore need to be studied as participants in broader behavioral systems (Rahwan et al., 2019).

These literatures support the precursor mechanisms. They do not establish the full MACS substrate. The theorem must therefore remain explicit about what is observed, what is partially supported, and what is genuinely forward-looking.

ABSTRACT

The Multi-Agent Cognitive Substrate Theorem (MACS) proposes that when heterogeneous cognitive participants—including humans, institutions, and autonomous computational systems—recurrently observe, classify, interpret, and modify one another's outputs within shared decision environments, epistemic stability becomes increasingly dependent on network-level interaction properties rather than solely on the quality of individual participants.

MACS formalizes three principal mechanisms: local drift, directional cross-agent drift transfer, and visibility divergence. Let $D_i(t)$ denote drift generated within participant i and $\phi_{i \rightarrow j}(t)$ denote distortion transmitted from participant i to participant j . Network drift is represented as:

$$D_N(t) = \sum_i D_i(t) + \sum_{\{i \neq j\}} \phi_{i \rightarrow j}(t).$$

MACS does not presume that cross-agent drift exceeds local drift across current systems. It introduces a testable network-dominance ratio:

$$\Phi_N(t) = \sum_{\{i \neq j\}} |\phi_{i \rightarrow j}(t)| / [\sum_i |D_i(t)| + \epsilon].$$

A network-dominant instability regime may be defined when Φ_N exceeds an empirically justified threshold. No universal threshold is declared.

MACS also introduces visibility divergence:

$$\Omega_V(t) = \sum_{\{i < j\}} w_{ij} d[V_i(t), V_j(t)],$$

where V_i is the representation available to participant i and d is a domain-appropriate divergence function. The theorem predicts that coordination risk increases when cross-agent transfer and visibility divergence rise while synchronization, grounding, and corrective damping remain weak.

MACS is therefore not a universal collapse theorem. High connectivity can generate distributed intelligence and resilience when grounding, interpretability, timing, and correction remain sufficiently coherent. The full substrate remains prospective, but the interaction mechanisms are empirically anchored in distributed cognition, human-AI collaboration, hybrid intelligence, platform-mediated coordination, multi-agent systems, and recursive sociotechnical feedback.

1. INTRODUCTION — WHEN THE AGENT IS NO LONGER THE SYSTEM

Earlier SR frameworks often treated a person, institution, platform, or AI system as a meaningful unit of analysis. That remains appropriate in many domains. MACS identifies the condition under which it becomes insufficient.

Consider a recurrent loop:

human behavior → institutional data → computational classification → institutional action → changed human behavior → new data.

No single element owns the entire causal process. Each participant contributes to the state that the next participant receives.

A more complex environment may include several models, organizations, platforms, regulators, users, and automated agents. Each sees a different representation of the system. Each acts at a different speed. Each may optimize a different objective. Each may feed outputs back into the others.

Under these conditions, the important analytical question changes.

Instead of asking:

Which agent failed?

MACS asks:

What interaction structure produced the observed state?

2. COGNITIVE PARTICIPATION — A FUNCTIONAL DEFINITION

MACS uses a constrained definition of cognitive participation.

A participant i is cognitively consequential to the network when it performs one or more functions that materially alter downstream epistemic or decision states, including:

observation;
classification;
ranking;
interpretation;
prioritization;
recommendation;
decision;
action selection;
feedback generation;
or modification of the informational environment used by other participants.

This definition does not imply consciousness, subjective experience, moral equivalence, or identical cognitive architecture across humans and machines.

It identifies functional participation.

Let $C_i(t)$ represent the decision-relevant cognitive state of participant i . The state can contain beliefs, classifications, scores, model outputs, institutional categories, priorities, or other representations that influence downstream action.

A MACS environment exists only when these states are relationally consequential:

$$\partial C_j(t+1)/\partial C_i(t) \neq 0$$

for a non-trivial subset of participant pairs.

The theorem therefore requires coupling, not mere coexistence.

3. LOCAL DRIFT AND NETWORK DRIFT

Each participant can generate internal drift.

$D_i(t)$ represents deviation within participant i from a relevant grounded baseline, objective, calibrated reference, or empirically justified expectation.

Examples include:

human interpretive drift;
institutional classification drift;
model distribution shift;
proxy distortion;
organizational policy drift;
or platform optimization drift.

If participants were independent, system drift could be approximated as:

$$D_{\text{local}}(t) = \sum_i D_i(t).$$

MACS becomes necessary when that additive model is inadequate because one participant's drift changes another participant's state.

Network drift is therefore:

$$D_N(t) = \sum_i D_i(t) + \sum_{\{i \neq j\}} \varphi_{i \rightarrow j}(t).$$

The second term is the MACS addition.

4. DIRECTIONAL DRIFT TRANSFER

$\varphi_{i \rightarrow j}(t)$ represents directional drift transfer: the amount of distortion, instability, misclassification, or grounding error transmitted from participant i into the state or behavior of participant j .

Transfer need not be symmetric.

$$\varphi_{i \rightarrow j} \neq \varphi_{j \rightarrow i}.$$

A machine ranking system can influence human behavior more strongly than individual users influence the model in a single period. A regulator can constrain a platform strongly while receiving only filtered information back. An institution can provide labels that become training data for models, while later model outputs reshape the institution's classifications.

Examples include:

model recommendations changing administrator behavior;
algorithmic ranking changing user attention;

user engagement changing future recommendation models;
institutional categories shaping training labels;
autonomous systems passing generated outputs to other autonomous systems;
or one platform's classification becoming an input to another organization's decision process.

Directional drift transfer makes error relational.

5. THE NETWORK-DOMINANCE RATIO

Whether cross-agent transfer exceeds local drift is an empirical question that must be tested in each multi-agent system.

MACS replaces it with:

$$\Phi_N(t) = \sum_{i \neq j} |\phi_{i \rightarrow j}(t)| / [\sum_i |D_i(t)| + \epsilon].$$

Interpretation:

$\Phi_N < 1$ — locally generated instability is larger than measured cross-agent transfer.

$\Phi_N \approx 1$ — local and interactional instability are comparable.

$\Phi_N > 1$ — measured interactional transfer exceeds local drift under the selected operationalization.

This ratio is not itself a universal law. It is a diagnostic quantity.

A domain must establish valid measures of D_i and $\phi_{i \rightarrow j}$ before Φ_N is meaningful.

A future claim that a system has entered a network-dominant MACS regime must therefore be demonstrated rather than asserted.

6. VISIBILITY BECOMES PLURAL

MACS establishes that visibility cannot be treated as a single system variable once participants operate through different representations.

Let:

$V_H(t)$ = human-layer visibility;

$V_I(t)$ = institutional visibility;

$V_M(t)$ = machine-layer visibility;
 $V_A(t)$ = autonomous-agent visibility;

with additional layers introduced as necessary.

These are not necessarily commensurable.

A person may observe lived consequences that do not appear in the institutional record. The institution may observe administrative categories invisible to the individual. A model may observe feature vectors and latent representations neither side can directly inspect. A second model may receive only the first model's output.

Thus:

$$V_H \neq V_I \neq V_M \neq V_A.$$

The problem is not simply incomplete visibility. It is representational divergence.

7. VISIBILITY DIVERGENCE

MACS introduces Visibility Divergence Ω_V :

$$\Omega_V(t) = \sum_{\{i < j\}} w_{ij} d[V_i(t), V_j(t)].$$

Here d is a domain-specific divergence function. It may represent disagreement in classification, discrepancy in predicted state, semantic distance, probability distribution divergence, or difference from a shared ground-truth reference.

The weights w_{ij} represent the causal importance of each pairwise discrepancy.

High Ω_V does not automatically produce failure. Diverse perspectives can improve collective judgment.

Risk increases when divergence is consequential and unreconciled.

A useful conditional proposition is:

$$\partial R_{\text{coord}} / \partial \Omega_V > 0$$

when reconciliation capacity is low.

This connects MACS to SR Ground-Truth Discrepancy. A network can contain several internally coherent but mutually incompatible representations of the same substrate.

8. SYNCHRONIZATION AS A CENTRAL VARIABLE

Synchronization is a central stabilizing variable in coupled multi-agent systems.

Let $S_N(t)$ represent network synchronization and reconciliation capacity.

$$S_N = f[G_C, I_C, T_C, R_C],$$

where:

G_C = grounding coherence across participants;

I_C = cross-agent interpretability;

T_C = temporal synchronization;

R_C = reconciliation and corrective capacity.

Synchronization does not mean identical beliefs or outputs. It means the network can identify, compare, and reconcile consequential differences before they propagate beyond recoverable bounds.

A system can therefore tolerate high diversity and even high local drift if S_N remains strong.

Conversely, even individually accurate participants can generate unstable collective behavior when timing, grounding, or reconciliation fails.

9. TEMPORAL ASYNCHRONY

MACS adds a problem that earlier institutional models often understate: participants operate at different speeds.

Humans may deliberate over minutes, hours, or days.

Institutions may update policies over months or years.

Models may classify in milliseconds.

Autonomous systems may exchange outputs continuously.

This creates temporal asymmetry.

Let τ_i denote the characteristic update interval of participant i .

Temporal divergence can be represented as:

$$\Omega_T = \text{Var}(\tau_1, \tau_2, \dots, \tau_n)$$

or through pairwise timing mismatch.

High Ω_T can matter when fast participants repeatedly alter the environment before slower participants can audit, contest, or correct them.

MACS therefore predicts that temporal synchronization is not merely an engineering concern. It becomes an epistemic governance variable.

10. NETWORK PROPAGATION

The network structure can be represented by a weighted adjacency matrix $W=[w_{ij}]$, where w_{ij} measures the influence of participant i on participant j .

Let D_t be a vector of participant-level drift states.

A simplified propagation model is:

$$D_{t+1} = (A+W)D_t + \epsilon_t,$$

where A captures internal persistence and W captures cross-participant propagation.

The spectral radius $\rho(A+W)$ provides one possible stability indicator.

If:

$$\rho(A+W) < 1,$$

small perturbations tend to decay in the linearized system.

If:

$$\rho(A+W)>1,$$

perturbations can amplify under that model.

This does not prove that real sociotechnical systems are linear. The formulation is a tractable starting point for network stress testing.

Its conceptual implication is powerful: instability can arise because of the topology and strength of interactions even when no single node is catastrophically unstable.

11. HARM UNDER MACS

MACS does not claim that all harm becomes network-distributed.

It introduces a network contribution to harm.

Let:

$$H_T = H_L + H_N,$$

where H_L is locally generated harm and H_N is harm amplified or transmitted through network interactions.

A richer formulation is:

$$H_N = f(D_N, \Omega_V, \Omega_T, S_N^{-1}, P_A),$$

where P_A represents propagation amplitude or reach.

Network-level harm can emerge through:

- cascading misclassification;
- recursive feedback;
- distributed false positives;
- distributed false negatives;
- cross-platform amplification;
- coordination failure;
- conflicting optimization targets;
- or delayed reconciliation between fast and slow participants.

The same network can also distribute correction.

Distributed human review, multiple models, institutional checks, and redundant sensing can dampen errors faster than a centralized system.

MACS is therefore symmetric in principle: networks can amplify both failure and correction.

12. STABLE MACS REGIMES

A stable MACS regime may emerge when:

- participants maintain distinct but reconcilable representations;
- important outputs are traceable across participant boundaries;
- ground-truth checks exist outside the recursive network;
- temporal mismatches are bounded;
- optimization objectives do not remain hidden from other consequential participants;
- errors can be propagated backward as well as forward;
- and no single high-centrality node can distort the network without detection.

Under these conditions, connectivity can create distributed intelligence.

The network may outperform isolated participants because errors are detected redundantly, expertise is complementary, and correction is shared.

13. UNSTABLE MACS REGIMES

An unstable regime may emerge when:

- cross-agent dependence rises;
- visibility divergence becomes consequential;
- fast automated participants recursively modify the environment;
- slower institutions cannot reconcile changes;
- models train on outputs generated by prior models;
- human behavior adapts to machine classifications;
- platforms optimize against incompatible objectives;
- or no participant retains an external grounding reference.

The danger is not simply bad AI.

It is recursive epistemic closure.

A network can become internally self-consistent while drifting away from the substrate it is supposed to represent.

This is where MACS intersects with zombie systems and Ground-Truth Discrepancy. A network may continue producing highly coordinated outputs while losing contact with lived or physical reality.

14. FORMAL THEOREM STATEMENT

Theorem — Multi-Agent Cognitive Substrate.

When heterogeneous cognitive participants recurrently observe, classify, interpret, and modify one another's outputs within a shared decision environment, system-level epistemic stability becomes increasingly dependent on interaction structure rather than solely on the reliability of individual participants. Under these conditions, drift can be generated locally and transferred directionally between participants; visibility can diverge across layers; timing can become asynchronous; and errors or corrections can propagate through the network. The probability of network-level instability increases when interactional transfer, visibility divergence, and temporal mismatch rise while grounding, interpretability, synchronization, and corrective damping remain weak.

Formally:

$$D_N = \sum_i D_i + \sum_{\{i \neq j\}} \phi_{i \rightarrow j}$$

$$\Phi_N = \sum_{\{i \neq j\}} |\phi_{i \rightarrow j}| / [\sum_i |D_i| + \epsilon]$$

$$\Omega_V = \sum_{\{i < j\}} w_{ij} d(V_i, V_j)$$

$$S_N = f(G_C, I_C, T_C, R_C)$$

$$R_{MACS} = f(\Phi_N, \Omega_V, \Omega_T, S_N^{-1}, P_A).$$

MACS does not require $\Phi_N > 1$, does not require a single shared cognitive state, and does not imply artificial consciousness.

15. EMPIRICAL PRECURSORS

Distributed cognition provides the oldest direct conceptual precursor. Hutchins (1995) demonstrated that cognitive work can be distributed across people, artifacts, representations, and coordinated procedures. Hollan, Hutchins, and Kirsh (2000) explicitly extended distributed cognition into human-computer interaction, emphasizing whole environments and coordination between people and technologies.

Hybrid intelligence provides a second precursor. Dellermann et al. (2019) describe configurations in which human and machine intelligence are combined to achieve outcomes neither component necessarily produces alone.

Human-AI teaming provides a third. Seeber et al. (2020) developed a research agenda around AI systems functioning as teammates rather than merely tools, explicitly identifying benefits and risks across machine, collaboration, and institutional dimensions.

Machine behaviour provides a fourth. Rahwan et al. (2019) argue that AI systems increasingly mediate human social, cultural, economic, and political interactions and should be studied behaviorally within those environments.

These literatures establish that cognition, decision-making, and behavior can be distributed across sociotechnical configurations. They do not establish the MACS claims about network-dominant drift, visibility divergence, or future autonomous multi-agent governance. Those propositions remain testable extensions.

16. EMPIRICAL STATUS

MACS should be classified as:

Forward substrate theorem with empirically supported precursor mechanisms; full network-distributed substrate not yet established.

Supported propositions:

Cognitive work can be distributed across people, artifacts, and technologies.

Human-AI collaboration can create joint decision structures not reducible to passive tool use.

AI systems already mediate human and institutional interactions.

Network structure can affect propagation, coordination, and collective outcomes.

Partially supported propositions:

Cross-agent feedback can create interactional error not attributable to one participant.

Visibility can diverge between humans, institutions, and machine systems.

Timing asymmetries can affect oversight and coordination.

Hybrid networks can either amplify or correct local error.

Prospective propositions:

Network-level interaction becomes more causally important than agent-local behavior across major social systems.

Cross-agent drift exceeds local drift in defined environments.

Autonomous systems routinely consume and modify one another's outputs in institutionally consequential networks.

Societal epistemic stability becomes meaningfully measurable as a network property.

17. FALSIFICATION CONDITIONS

Falsifier 1 — Interaction adds no explanatory power.

If participant-level variables predict system outcomes as well as or better than models including cross-agent transfer, MACS network terms are unnecessary in that domain.

Falsifier 2 — Drift transfer is negligible.

If $\phi_{i \rightarrow j}$ is consistently small relative to D_i across mature coupled systems, claims of network-distributed drift must be narrowed.

Falsifier 3 — Visibility divergence does not matter.

If Ω_V has no relationship to coordination error, disagreement persistence, or correction latency, visibility divergence should not remain a central variable.

Falsifier 4 — Synchronization is not predictive.

If S_N adds no explanatory value for stability or recovery under comparable network conditions, the synchronization proposition fails.

Falsifier 5 — Network effects remain local.

If observed cross-agent effects do not propagate beyond small dyadic interactions and do not generate higher-order system behavior, MACS should remain a local interaction framework rather than a substrate theorem.

Falsifier 6 — Connectivity consistently improves outcomes.

If higher cross-agent coupling under realistic conditions reliably improves grounding, correction, and stability without generating interactional failure modes, MACS risk claims must be substantially reduced.

18. RELATIONSHIP TO EXISTING LITERATURE

MACS overlaps with distributed cognition, collective intelligence, multi-agent systems, network science, human-AI collaboration, sociotechnical systems, cybernetics, and complex systems theory.

Distributed cognition already demonstrates that cognition can be distributed across people and artifacts.

Multi-agent systems research studies coordination, cooperation, competition, communication, and emergent behavior among interacting agents.

Network science studies how topology, centrality, coupling, and diffusion shape system behavior.

Human-AI teaming studies joint performance, trust, transparency, role allocation, and collaboration.

MACS does not claim priority over those literatures.

Its distinct contribution is to integrate them with SR variables of drift, visibility, grounding discrepancy, propagation, and harm inside a single substrate-transition model.

19. RELATIONSHIP TO SR

MACS is where several SR operators become explicitly networked.

Propagation becomes directional $\phi_{i \rightarrow j}$ rather than a single scalar transfer term.

Visibility Lag becomes layer-specific and temporally asynchronous.

Ground-Truth Discrepancy becomes multi-representational: different participants can diverge from the same substrate in different ways.

Silo Exposure Bias can become topological when one participant sees only the outputs of another and mistakes that partial representation for the whole system.

Zombie Metrics can circulate recursively between institutions and models, acquiring authority through repetition rather than renewed grounding.

Algorithmic inheritance can become network inheritance when one system's classifications seed another system's inputs.

MACS therefore transforms SR from an institutional diagnostic architecture into a network diagnostic architecture without abandoning the institutional layer.

20. THE TRILOGY AS A SINGLE ARCHITECTURE

The trilogy is:

CRI — Persistent Observation.

AIS — Delegated Decision.

MACS — Network-Dependent Cognition.

The sequence can be represented as:

increasingly persistent sensing → computational interpretation → consequential institutional delegation → recurrent cross-agent dependence → network-level epistemic behavior.

Each transition is analytically separable.

CRI can exist without AIS.

AIS can exist without full MACS.

MACS requires recurrent coupling, not simply the presence of AI.

The trilogy therefore describes a structural progression that may unfold unevenly across sectors and countries rather than a deterministic timeline.

21. WHY MACS MATTERS FOR INSTITUTIONS

MACS is the theorem that most directly challenges the assumption that institutions remain the sole or even primary locus of coordinated social cognition.

Formal authority may remain institutional.

Effective observation may be distributed across sensors, platforms, models, and people.

Effective judgment may be distributed across administrators and computational systems.

Effective propagation may occur across networks no single institution fully controls.

Under those conditions, an institution can retain legal authority while losing epistemic centrality.

That is not institutional disappearance.

It is a possible divergence between formal sovereignty and functional cognitive position.

This distinction should become central to future SR planetary work. A state may remain legally intact while increasingly depending on external or semi-autonomous systems for sensing, prediction, prioritization, infrastructure coordination, and policy execution. The relevant question becomes whether it can still understand, audit, and redirect the network on which its effective governance depends.

22. CANONICAL PARAGRAPH

The Multi-Agent Cognitive Substrate emerges when humans, institutions, and autonomous computational systems become recurrently coupled participants in shared epistemic and decision environments. MACS does not claim that cognition becomes a single mind or that machines possess human consciousness. It defines cognitive participation functionally: participants matter when their outputs alter what others observe, classify, prioritize, decide, or learn. Under these conditions, drift can arise locally and transfer directionally across participants, visibility can diverge between layers, timing can become asynchronous, and stability can depend on network synchronization, grounding coherence, interpretability, and corrective damping. MACS therefore formalizes the transition from agent-centered diagnosis to network-centered diagnosis: interaction itself becomes part of the causal substrate.

23. CONCLUSION

MACS establishes a bounded theorem of network-dependent cognition without presuming that network effects dominate every coupled system.

Cross-agent drift may exceed local drift in some regimes, but that relationship must be measured. Distributed cognition does not imply artificial consciousness, and network-level instability does not necessarily replace institution-specific failure.

The rigorous core is relational and testable.

When decision-relevant participants recursively alter one another's states, the behavior of the system can depend on the interaction network in ways that are not recoverable from isolated-agent analysis alone.

This proposition is already supported in principle by distributed cognition, hybrid intelligence, human-AI teaming, network science, and multi-agent research.

The full MACS substrate remains prospective.

Its canonical transition is:

Network-Dependent Cognition.

The trilogy therefore closes as:

CRI — reality becomes increasingly persistently observable.

AIS — computational interpretation becomes increasingly consequential to institutional judgment.

MACS — epistemic behavior becomes increasingly dependent on the network connecting humans, institutions, and autonomous systems.

The future SR diagnostic unit is therefore not always the person, the institution, or the machine.

Sometimes it is the coupled network.

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